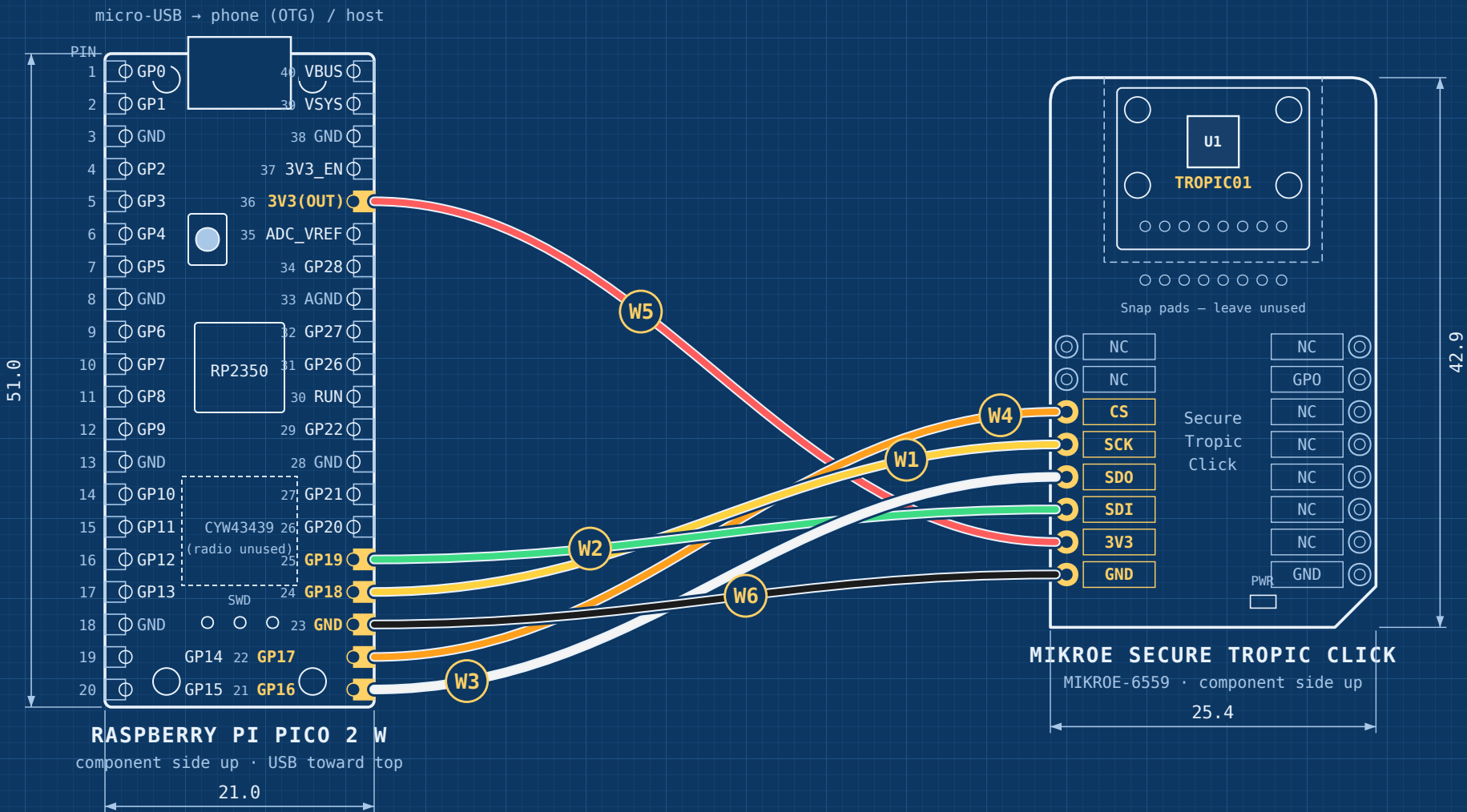


SHEET 1 · WIRING & PINOUT

Six wires, soldered directly into the plated holes. No headers, no other parts.



HARNESS / NET LIST

WIRE	SIGNAL	PICO 2 W	PIN	CLICK	DIR	COLOUR*
W1	SPI SCK	GP18	24	SCK	Pico → SE	yellow
W2	SPI MOSI	GP19	25	SDI	Pico → SE	green
W3	SPI MISO	GP16	21	SD0	SE → Pico	white
W4	CHIP SELECT	GP17	22	CS	Pico → SE	orange
W5	+3.3 V	3V3(OUT)	36	3V3	supply	red
W6	GROUND	GND	23 / 38	GND	return	black

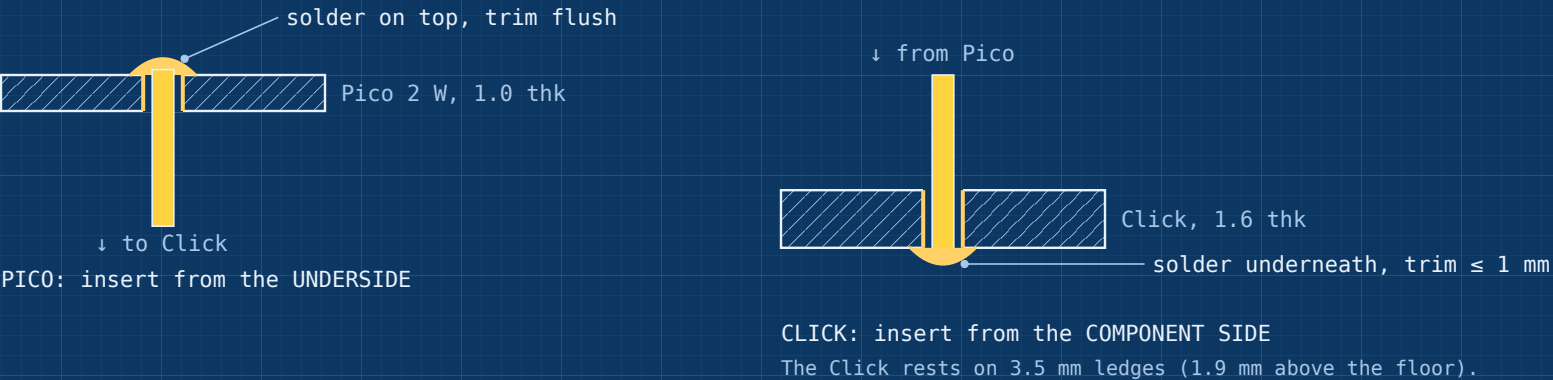
BUS PARAMETERS (fixed in firmware)

Peripheral	RP2350 SPI0 · 8-bit frames
Mode / clock	MODE 0 (CPOL 0, CPHA 0) · 1 MHz
Chip select	GP17 as push-pull GPIO, driven by firmware
Logic / supply	3.3 V only · Pico 3V3(OUT) pin 36
Source of truth	crates/dsm-anchor-pico/src/main.rs crates/dsm-anchor-secure-monitor/src/tropic.rs

NOTES

- 3.3 V ONLY. Never connect VBUS or VSYS (5 V) to the Click.
- No pin headers. Solder each wire straight into the plated hole on both boards – headers will not fit the pocket case.
- Leave GP0, every NC pad and the Snap pads unconnected.
- Do not snap off the Click Snap module; the case is sized for the whole Click.
- 28–30 AWG stranded silicone wire, 60–80 mm per lead.
- *Colours are a suggestion; signal names govern.
- Boards are drawn side by side for clarity. Installed, the Click lies under the Pico (see Sheet 2).

DETAIL A · WIRE ENTRY (section, 10:1 approx.)



LEGEND

- pad used by the harness
- pad left unconnected
- W wire number (see net list)

SE = secure element (TROPIC01)
Component outlines on both boards are indicative.
Pin names and pad positions are exact.

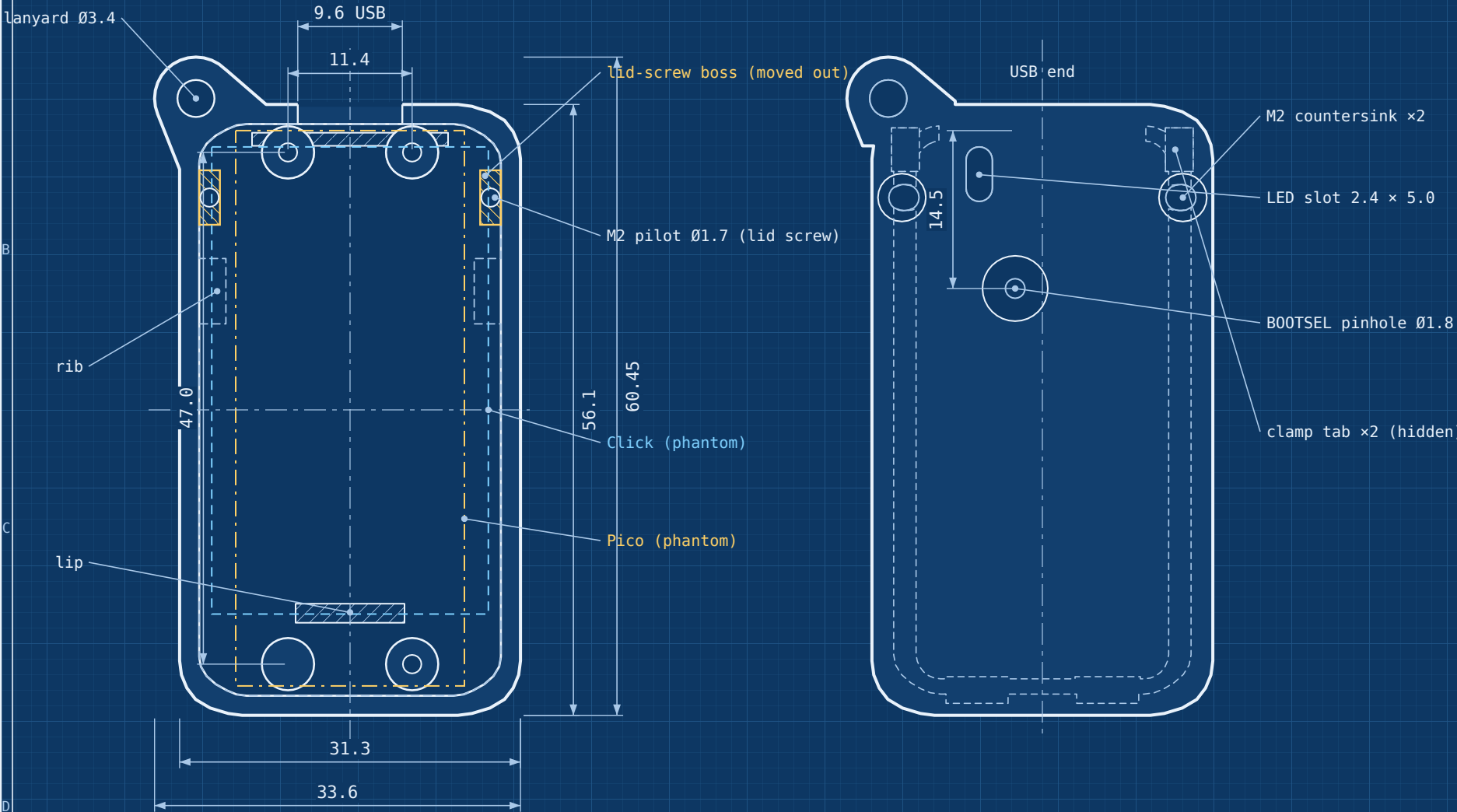
DSM · OFFLINE ANCHOR APPLIANCE
SHEET 1 – WIRING & PINOUT

PROJECT	deterministicstatemachine/dsm
ASSEMBLY	Pico 2 W (RP2350) + Secure Tropic Click (TROPIC01)
ENCLOSURE	Pocket case REV E · base.stl / lid.stl
UNITS	mm · drawings not to scale unless noted

SHEET	1 / 2
REV	E
DATE	2026-10-02
LICENSE	MIT/Apache-2.0

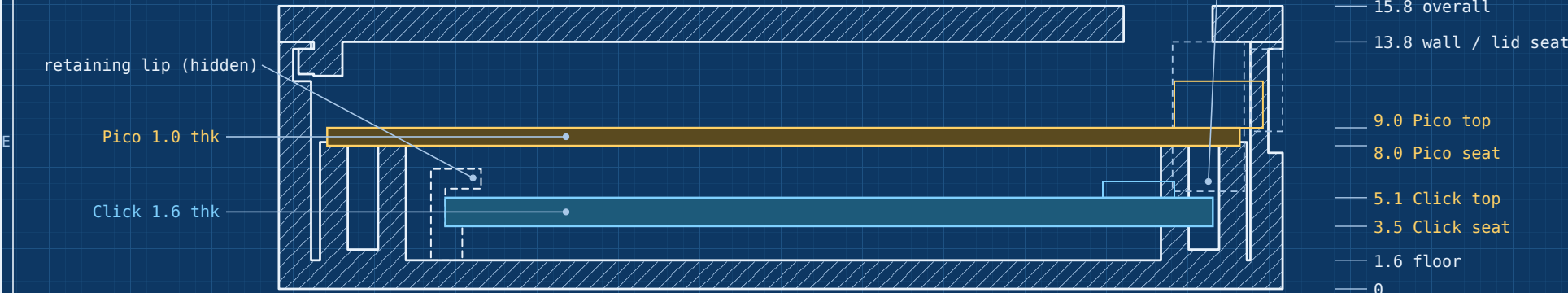
SHEET 2 · ENCLOSURE & STACK-UP

Pocket case REV E. Geometry traced from base.stl / lid.stl. Dimensions in mm.



VIEW A · BASE, TOP
phantom: Pico (yellow), Click (blue)

VIEW B · LID, TOP (as installed)
hidden lines: skirt, clamp tabs
clamp tab, tip at 5.45 (hidden)



SECTION C-C · through the standoffs (x = -5.7), USB end right

BILL OF MATERIALS

ITEM	QTY	DESCRIPTION
1	1	Raspberry Pi Pico 2 W (RP2350), no headers
2	1	MIKROE-6559 Secure Tropic Click (TROPIC01)
3	6	Wire, 28-30 AWG stranded silicone, 60-80 mm
4	4	Screw M2 × 5, self-tapping (Pico → standoffs)
5	2	Screw M2 × 6, countersunk self-tapping (lid)
6	1	base.stl – PETG (PLA ok)
7	1	lid.stl – PETG (PLA ok)
8	1	USB cable: micro-USB to phone (OTG) or host

PRINT SETTINGS

PETG preferred, PLA fine · 0.2 mm layers · 3 perimeters
25 % infill · NO supports · no brim · ~35 g total
Both parts print flat face down, already oriented
base ≈ 2 h · lid ≈ 1 h

ASSEMBLY SEQUENCE

- 1 Print base + lid. Test-fit the bare Pico on the four standoffs.
- 2 Wire the Click: six leads in from the component side, solder underneath, trim ≤ 1 mm (Sheet 1, Detail A).
- 3 Lay the Click component-up on the ribs, TROPIC01 end toward USB, far end tucked under the retaining lip.
- 4 Pass the leads up; solder them into the Pico from the underside (pins 21, 22, 23, 24, 25, 36). Trim flush.
- 5 Seat the Pico, USB into the opening. Screw down (4× M2×5).
- 6 Flash + self-test (README) BEFORE closing the lid.
- 7 Fit the lid: the tabs clamp the Click. 2× M2×6 countersunk.

NOTES

1. Pico sits component side UP on the standoffs (seat z = 8.0).
2. The Click lies in the 6.4 mm bay under the Pico (seat z = 3.5).
3. REV E: side walls, lid-screw bosses and lid countersinks moved out 0.75 mm per side so the Pico clears the bosses (view A).
4. Section C-C is traced from the STLs; components are phantoms.

DSM · OFFLINE ANCHOR APPLIANCE
SHEET 2 – ENCLOSURE & STACK-UP

PROJECT	deterministicstatemachine/dsm
ASSEMBLY	Pico 2 W (RP2350) + Secure Tropic Click (TROPIC01)
ENCLOSURE	Pocket case REV E · base.stl / lid.stl
UNITS	mm · drawings not to scale unless noted

SHEET	2 / 2
REV	E
DATE	2026-10-02
LICENSE	MIT/Apache-2.0